PheroSAM-3D Full-Scope Implementation Plan

Oct 30, 2026 · @Omar Hazem

Overview

This plan builds everything in the proposal: all five contributions, all six research questions and all four datasets, within the course deadlines (weeks 7, 11 and 13). It only works with three parallel workstreams, a working pipeline by the end of week 9, and a clear cut order if anything slips (last section).

It is about twice the work of the single-organ plan in the other doc. Treat the single-organ plan as the safe floor: if week 9 ends late, fall back to it.

Dates assume week 7 ends on 31 October 2026; replace them with your course dates.

Course phase	Week	Assumed due	Must show
Phase 1: proposal + initial implementation	7	this week	Dataset works, baseline works, NIC designed and partly running
Phase 2: optimization + evaluation	11	about 28 Nov	Complete pipeline with measurable results on all datasets
Phase 3: code, report/presentation, paper (.docx + .pdf)	13	about 12 Dec	Everything, written up

The five contributions, all in scope:

1. Hybrid search: MMAS chooses a box per slice, ACO_R refines it
2. DP-seeded ACO, with the "why not DP" analysis
3. Competing colonies for multi-organ consistency, with anatomical rules
4. Doctor corrections that spread through the volume
5. Model-agnostic layer: the colony picks MedSAM2 keyframes (plug-in mode)

Plus the minor component: pheromone memory across hospitals.

Datasets, all four:

Dataset	Use	Source domain (proposer trained here)	Test
Multi-site prostate MRI	Main single-organ study, memory stream	RUNMC	Other 5 sites
M&Ms cardiac cine MRI	Vendor shift; 3 nested structures (LV, RV, myocardium)	One vendor	Other vendors
AMOS 2022	15 organs, competing colonies; CT→MRI shift	AMOS CT training set	About 50 CT + 20 MRI test volumes
MSD Liver (Task 3)	Honest tumor case: liver + tumor reported separately	MSD training split	Held-out split

Test sets are subsampled to fit the GPU budget; numbers above are starting targets.

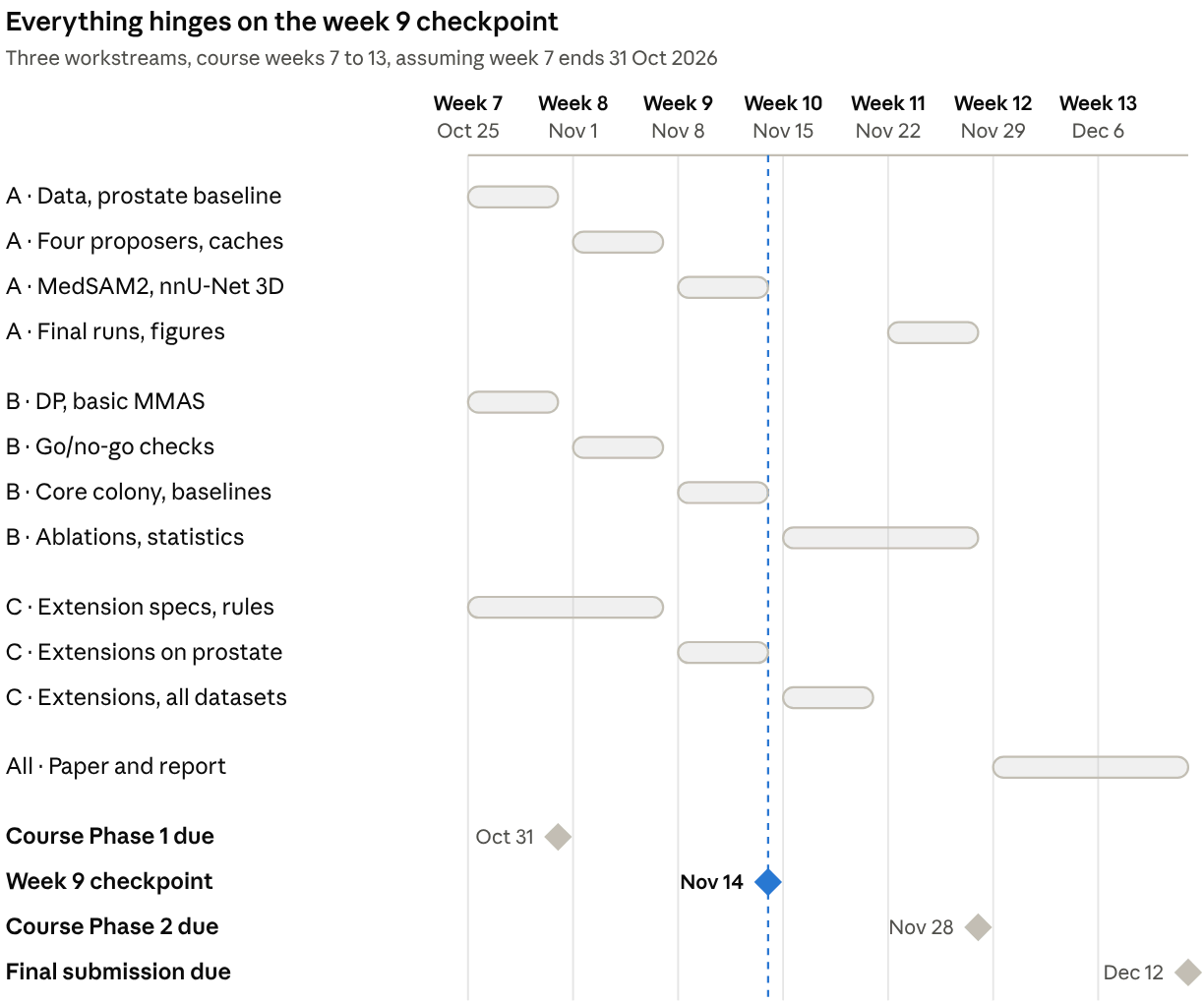
Research questions and where each is answered:

RQ	Question (short)	Answered by
1	Does the hybrid colony beat all baselines and exceed the best proposer box?	Week 9–11 main comparison
2	When is DP not enough, and how much does seeding help?	Global-weight sweep, seeding ablation
3	Do competing colonies reduce overlap and anatomy errors?	AMOS + M&Ms multi-structure experiments
4	How many corrections reach a target Dice?	Correction curves, all datasets
5	Does the label-free score track Dice; do flags find wrong slices?	Week 8 checks + flag evaluation
6	Can the colony improve MedSAM2 by choosing keyframes?	Plug-in experiment

Three workstreams (for 3 people; with 2, the optimization owner also takes C):

Workstream	Owns
A. Data and models	All 4 datasets, preprocessing, 4 proposers, MedSAM caches, shape autoencoders, MedSAM2 runs, metrics
B. Core optimization	Fitness, MMAS, DP seed, ACO_R, all baseline optimizers, tuning, statistics
C. Extensions	Competing colonies + rules, corrections + flags, pheromone memory, MedSAM2 plug-in

Roadmap



full-scope roadmap · 3 workstreams, 4 milestones

If the core colony is not running on all four datasets by the week 9 checkpoint, switch to the single-organ plan that day; the extensions in workstream C cannot be finished in week 10 otherwise.

Week 7 — Course Phase 1 submission (due this week)

The Phase 1 code runs on prostate only; the report presents the full design. Meanwhile workstream A starts downloading the other three datasets, because data access is the slowest step.

Workstream A

1. Download all four datasets; register where needed (M&Ms and AMOS require it). Record source and version in `DATA.md`.
2. Prostate loader; MedSAM preprocessing copied from MedSAM's own inference code; cache encoder embeddings for 10 test volumes.
3. Quick proposer: a small 2D U-Net trained on RUNMC for about an hour. If there is no GPU time, use jittered true boxes and say so in the report.
4. Baselines: MedSAM best box per slice; greedy propagation; 3D Dice per volume.

Workstream B

1. Candidates (3–5 per slice + no-object), MedSAM masks and confidence.
2. Fitness with Conf + Smooth; DP (Viterbi); basic MMAS (discrete only, 20 ants, 30 rounds).
3. Results table on 10 volumes: best box, greedy, DP, MMAS, oracle boxes.

Workstream C

1. Write the design of every extension as short specs: competing-colony schedule and repulsion, anatomical rules per dataset, correction loop, memory fingerprint and prior, MedSAM2 plug-in graph. These go into the report's sections 6–8.
2. Draft the anatomical rule lists for AMOS and M&Ms (to be validated on training masks in week 8).

Report (6–8 pages) + code: proposal sections 1–7 shortened and fixed (the "arg arg" equations, placeholder references); section 8 = what runs; sections 9–10 = the table and one figure. State that MMAS not yet beating DP is expected with local terms only.

Done when:

- [ ] All four datasets downloaded (or access requested)

- [ ] Prostate baseline, DP and basic MMAS results on 10 volumes
- [ ] Extension specs written
- [ ] Phase 1 report and repo submitted

Week 8 — Four datasets, proposers, caches, go/no-go checks

Goal: every dataset is cached and checked by the end of week 8, so weeks 9–10 run only on cached data.

Workstream A — data and models

1. Leakage check for all four datasets against MedSAM's training-data list (prostate first, then AMOS, which is reported as overlapping). Decide per dataset: use, exclude overlapping cases, or report.
2. Preprocessing per dataset:

Dataset	Preprocessing notes
Prostate MRI	Percentile normalization; common in-plane spacing; axial slices
M&Ms	End-diastole and end-systole frames as separate volumes; short-axis slices; percentile normalization. Few slices per volume (about 10), so smoothness matters less here
AMOS CT	Abdominal CT window; common spacing; axial slices
AMOS MRI	Percentile normalization; used only as unseen test domain
MSD Liver	CT window; liver and tumor kept as two structures

3. Train one 2D nnU-Net v2 proposer per dataset on its source domain (4 models; one fold each; start all on day 1, on separate GPU accounts if possible).
4. Candidates and MedSAM caches for all test volumes: embeddings (float16), candidate masks, predicted IoU, stability score, candidate type labels.
5. Shape autoencoders: one per structure type (prostate, LV, RV, myocardium, liver, and the AMOS organs grouped by size if time is short). Fallback: simple priors (volume range, compactness, slice-area profile).

6. Image fingerprint for memory: intensity-histogram statistics + mean/std of nnU-Net encoder features, saved per volume.

Workstream B — checks (on each dataset's validation split)

Check	Pass if
Headroom: oracle selection vs best-confidence box	About 3 Dice points or more
Candidate recall: slices with a candidate of IoU ≥ 0.7	About 80% or more
Confidence: predicted IoU vs true slice Dice	Spearman about 0.4 or more; else use stability score
Fitness: F vs true 3D Dice over about 200 random paths	Median Spearman about 0.5 or more

A dataset that fails headroom stays in the paper as a reported limitation, not a dropped result.

Workstream C — rules and specs

1. Validate anatomical rules on training masks; keep a rule only if it holds on at least 99% of cases. Write rules in patient coordinates.
   a. AMOS examples: liver centroid on the patient's right of the spleen; top of the liver above the top of the right kidney
   b. M&Ms: myocardium surrounds the LV; RV lies beside the LV, not inside it
   c. MSD Liver: tumor inside the liver
2. Build the MedSAM2 baseline runner (box on center slice, bidirectional propagation) and time one volume per dataset. This sets the plug-in budget.

Done when:

- [ ] Leakage decisions for all four datasets
- [ ] Four proposers trained; all caches complete
- [ ] Go/no-go table per dataset
- [ ] Rule lists validated; MedSAM2 runner timed

Week 9 — Core hybrid colony and all baselines

Goal: contributions 1 and 2 complete and running on all four datasets, single structure at a time. This is the checkpoint for the whole plan: if it is not done by the end of week 9, switch to the single-organ plan.

Workstream B — the core colony

1. One interface for every optimizer: `optimize(cache, fitness, budget, seed, fixed_nodes) -> path`. The `fixed_nodes` argument is there from the start, so corrections plug in later without changes.
2. Full fitness: Conf + Smooth + Shape + Agree − Pen.
3. MMAS: pheromone on edges incl. no-object; best-ant deposit (iteration-best, then global-best); τ_min and τ_max from the standard formulas; stop after 10 rounds without improvement.
4. DP seed: Viterbi on Conf + Smooth; DP edges start at τ_max, all others at 0.5 × τ_max.
5. ACO_R refinement: offsets (dx, dy, dw, dh) as fractions of box size, about ±15%; archive of the best 10 per slice; spread ξ = 0.85; boxes rounded to a 2-pixel grid and cached; new boxes decoded in one GPU batch per round.
6. Tuning on validation splits only (20–30 Optuna trials); freeze in config.

Workstream A — baselines that need models

1. MedSAM with nnU-Net boxes, slice by slice
2. MedSAM2 with a center-slice box
3. nnU-Net 3D as a supervised reference (prostate and MSD Liver only if GPU time is short)
4. Oracle MedSAM (true boxes)

Workstream B or C — optimizer baselines, same budget (600 fitness evaluations per structure per volume)

Best box per slice, greedy propagation, DP, random search, GA, PSO, simulated annealing, beam search. Same tuning effort as ACO.

Ablations (start runs at the end of the week): Ant System vs MMAS; ±DP seed; ±ACO_R; ±Shape term; global-weight sweep for DP vs DP-seeded ACO.

Done when:

- [ ] Full hybrid colony runs on all four datasets
- [ ] All baselines run under the same budget

- [ ] First results table (1 seed) per dataset
- [ ] Decision recorded: stay on full scope or switch to the single-organ plan

Week 10 — Competing colonies, corrections, memory, MedSAM2 plug-in

Goal: contributions 3, 4 and 5 plus pheromone memory are complete. Workstream C starts coding them in week 9 on the prostate cache, so week 10 is integration and runs, not first drafts. Code is tagged `v1.0` at the end of this week; after that, bug fixes only.

1. Competing colonies (contribution 3)

1. One colony per structure: 15 for AMOS (the two kidneys and the two adrenal glands are already separate labels), 3 for M&Ms (LV, RV, myocardium), 2 for MSD Liver (liver, tumor).
2. Colonies take turns, one round each; when one colony scores its ants, the others' current best masks are held fixed.
3. Joint fitness: sum of per-structure F − μ × overlapping voxels − ν × broken rules.
4. Repulsion: lower the confidence term of boxes overlapping another colony's best mask (proposal formula, ζ = 2).
5. Allowed overlaps: myocardium–LV in M&Ms and tumor–liver in MSD are containment, not conflict. Encode them as "contains" rules, not as overlap penalties.
6. Comparisons: independent colonies; independent colonies + post-hoc overlap fix (voxel goes to the more confident structure); competing without repulsion; full competing; GA and PSO with one joint solution over all structures.

2. Flags and corrections (contribution 4)

1. Pheromone entropy per slice; flag the top 3 per volume and structure.
2. Simulated doctor fixes the worst flagged slice with its true box → fixed node (edges clamped at τ_max, other options removed) → about 10 more rounds.
3. Run 1, 2, 3, 5 and 10 corrections; compare with greedy propagation, DP with fixed nodes, and MedSAM2 with the corrected slice added as a prompt.

3. Pheromone memory (minor component)

1. After each volume, store a compact prior: 10 bins of relative slice position × pheromone share per candidate type, plus the volume's fingerprint.

2. New volume: find the nearest stored fingerprint; if close enough, start from $\tau_init = (1 - \lambda)\ \tau_max + \lambda \times memory$, $\lambda = 0.5$; otherwise start flat and create a new entry. Update the matched entry by moving average.
3. Stream experiment on prostate: test sites in the order B → C → D → B → C → D. Measure rounds to converge and Dice on the first vs second visit, with and without memory.
4. Repeat on M&Ms vendors if time allows.

4. MedSAM2 plug-in mode (contribution 5)

1. New graph: nodes are (slice, candidate box) pairs; an ant picks 1–3 keyframes with their boxes.
2. MedSAM2 propagates from them; the same label-free F scores the volume.
3. Fewer ants and rounds (for example 8 ants × 10 rounds), because each evaluation reruns propagation; run on GPU.
4. Compare with MedSAM2 center-slice box and with 3 evenly spaced keyframes, at matched numbers of keyframes.
5. Run on prostate and a 20-volume AMOS subset; extend only if the GPU budget allows.

Done when:

- [ ] Each extension produces a CSV on at least one dataset
- [ ] Competing colonies run on AMOS CT and M&Ms
- [ ] Code tagged `v1.0`

Week 11 — Course Phase 2 submission (about 28 Nov)

Goal: all experiments rerun from `v1.0` with 3 seeds, one table or graph per research question, an error analysis per dataset, and the Phase 2 report submitted.

Runs (book GPU time in advance; split across accounts by dataset):

Experiment	Datasets	Methods
Main comparison (RQ1)	All four	Hybrid colony vs all baselines, equal budget
Ablations + global-weight sweep (RQ2)	Prostate, AMOS CT	AS vs MMAS, ±DP seed, ±ACO_R, ±Shape

Experiment	Datasets	Methods
Multi-structure (RQ3)	AMOS CT, M&Ms, MSD Liver	Independent, post-hoc fix, competing ± repulsion, joint GA/PSO
Corrections (RQ4)	All four	Colony vs greedy vs DP vs MedSAM2
Score and flags (RQ5)	All four	Spearman F vs Dice; flag catch rate vs confidence and random
Plug-in (RQ6)	Prostate, AMOS subset	Colony keyframes vs center slice vs evenly spaced
Memory stream	Prostate (M&Ms if time)	With vs without memory
Domain shift	AMOS CT → MRI; M&Ms vendors; prostate sites	Same methods, results per domain

Statistics: Wilcoxon signed-rank per volume, Holm correction, mean ± std over 3 seeds.

Graphs and tables:

1. Main results table per dataset: Dice, HD95, slice consistency (tumors reported separately)
2. Convergence: fitness vs evaluations, all optimizers
3. DP vs DP-seeded ACO gap vs global-term weight
4. Share of slices where refinement beats every candidate box
5. Overlap voxels and rule violations, multi-structure table
6. Dice vs number of corrections
7. Fitness vs true Dice scatter; flag catch rate
8. MedSAM2 keyframe comparison
9. Memory: rounds to converge and Dice, first vs second visit
10. Cost: seconds per volume, decoder calls, GPU memory
11. Qualitative figure: greedy vs DP vs colony vs competing colonies, with the pheromone-entropy map

Error analysis:

1. Results by domain (site, vendor, CT vs MRI) and by slice position (organ ends vs middle)

2. Organs vs tumors: where the shape prior stops helping
3. The 10 worst volumes per dataset sorted into causes: proposer missed the structure; MedSAM mask leaks; wrong no-object at the ends; fitness preferred a wrong path; competing colonies pushed a correct mask away

Done when:

- [ ] Every row of the runs table has a CSV
- [ ] Graphs and tables rebuilt by script
- [ ] Error analysis with examples
- [ ] Phase 2 report and updated code submitted

Weeks 12–13 — Writing and final submission (about 12 Dec)

Goal: a full paper draft by the end of week 12; week 13 is review, packaging and submission only. With five contributions, the paper risks being a list; organize Results by research question, not by component.

Week 12 — paper draft (course structure, written in this order)

1. Proposed Method: slice graph, fitness, hybrid MMAS + ACO_R, DP seeding, competing colonies, corrections, memory, plug-in mode. Written from the code.
2. Materials and Methods: four datasets, domain splits, leakage handling, proposers, MedSAM and MedSAM2, baselines, equal budget, metrics, statistics.
3. Experimental Results: one subsection per research question (RQ1–RQ6) plus memory.
4. Discussion: what worked, the tumor case, domain-shift findings, the error analysis; limitations (simulated doctor, possible MedSAM data overlap, compute cost).
5. Related Work, Introduction, Conclusion; Abstract and Keywords last.
6. References checked against the real papers.
7. Final report or slide deck (about 15–18 slides) in the course's section order, reusing the paper's figures.
8. Send the draft to your instructor or a classmate by Thursday.

Week 13 — final packages

A. Source code, one folder or script per course item with a README command for each:

Course item	In this project
Data preprocessing	Loaders and preprocessing for all four datasets
Baseline model	MedSAM per slice, greedy propagation, MedSAM2, nnU-Net 3D, oracle
NIC algorithm	MMAS + DP seed + ACO_R, competing colonies, memory, plug-in mode; GA, PSO, SA, beam search
NIC-AI integration	Candidates → MedSAM / MedSAM2 → fitness → ants
Training	Four nnU-Net proposers, shape autoencoders
Testing	Running every method on every test domain
Evaluation	Metrics and statistics
Visualization	Every figure and table, rebuilt by script

B. Final report / presentation in the course's order: Introduction through Future work.

C. Research paper as .docx and .pdf, the PDF exported from the same Word file; check equations, figures and references after export.

Done when:

- [ ] Fresh clone of the repo reproduces the main table
- [ ] Every number in the paper matches a CSV
- [ ] Report or slides submitted
- [ ] Paper submitted as .docx and .pdf

Cut order and risks

If the schedule slips, cut from the top of this list down, one item at a time, and move each cut item to Future work. Never cut the protected core.

Protected core (never cut): hybrid colony with DP seed and ACO_R on prostate; equal-budget baselines; flags and corrections; error analysis. Together these are the single-organ plan and already satisfy every course deliverable.

Cut order:

1. MedSAM2 plug-in beyond prostate
2. Memory stream on M&Ms
3. nnU-Net 3D reference (keep oracle boxes as the upper bound)
4. AMOS MRI as a test domain
5. AMOS reduced from 15 organs to 5 (liver, spleen, both kidneys, stomach)
6. M&Ms cardiac
7. MSD Liver tumors
8. Pheromone memory entirely
9. MedSAM2 plug-in entirely
10. Competing colonies reduced to 3 AMOS organs; at this point you are on the single-organ plan

Risks:

Risk	Early warning	Response
Dataset access delayed (M&Ms, AMOS registration)	Not downloaded by end of week 7	Start with whatever arrived; drop the late one via the cut order
Four proposers too slow to train	Not finished by mid-week 8	Shorter-epoch trainer; reuse the week 7 small U-Net for prostate
GPU quota runs out	Week 8 timing × planned runs exceeds quota	Subsample test sets further; fewer ants and rounds for every method equally
Core colony late	Week 9 checkpoint missed	Switch to the single-organ plan the same day
Fitness does not track Dice on some dataset	Week 8 check fails	Report that dataset as a limitation; keep it in the paper
Competing colonies hurt containment structures	Myocardium or tumor Dice drops	Treat containment as rules, not overlap penalties (week 10, item 1.5)
Paper becomes a list of components	Week 12 draft review	Organize around the RQs; move minor results to an appendix